# Unified VDS/DVP/BH Number System Derivation with Ramanujan Acceleration

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## PAPER_1108: Unified VDS/DVP/BH Number System Derivation with Ramanujan Acceleration

### Abstract

We present a unified derivation engine for the three UQFF number systems — Vacuum Density Series (VDS), Dipole Vortex Primes (DVP), and Buoyancy Harmonics (BH) — with Ramanujan acceleration and cross-validation. These three number systems collectively govern the vacuum structure, vortex quantization, and charge-reactivity gravity ($U_{g2}$) of the UQFF framework.

### 1. VDS: Vacuum Density Series

#### Definition

$$\text{VDS} = \sum_{n=1}^{\infty} \frac{[\text{SSq}]^n}{n^{26}} = \text{Li}_{26}([\text{SSq}])$$

where $\text{Li}_{26}$ is the polylogarithm of order 26.

#### Ramanujan Acceleration

The convergence is accelerated by the factor $R_n^{(26,3)}$:

$$S_{26}^{(3)}([\text{SSq}]) = \sum_{n=1}^{\infty} \frac{[\text{SSq}]^n}{n^{26}} \cdot R_n^{(26,3)}$$

$$R_n^{(26,3)} = \frac{(2\pi)^{n/6}}{n!} \left[ 1 + \sum_{m=1}^{3} \frac{1}{n^{26m}} \sum_{j=1}^{26} (-1)^{j+1} \binom{26}{j} \frac{(26-j)!}{n^j} \right]$$

This achieves 60+ digit precision in $\leq 50$ terms.

#### Physical Meaning

VDS encodes the **vacuum density normalization** — the suppression of the cosmological vacuum energy through the 26-layer compactification structure. $[\text{SSq}] = 0.57$ ensures convergence and bounds the vacuum from above.

## 2. DVP: Dipole Vortex Primes

### Definition

For primes $p > 26$:

$$a(p) = \frac{[\mathrm{SSq}]^{\pi(p)}}{p^{26}}$$

where $\pi(p)$ is the prime-counting function.

### Key Values

| $p$ | $\pi(p)$ | $a(p)$ |
|---|---|---|
| 29 | 10 | $\approx 3.62 \times 10^{-42}$ |
| 31 | 11 | $\approx 4.92 \times 10^{-43}$ |
| 37 | 12 | $\approx 1.80 \times 10^{-48}$ |
| 41 | 13 | $\approx 2.60 \times 10^{-50}$ |
| 113 | 30 | $\approx 10^{-83}$ |

### Physical Meaning

DVP encodes **vortex quantization** — the proplyd orbital radius $r_q \approx 0.0973$ AU and Navier-Stokes bounded vorticity. The prime-indexed structure ensures each vortex mode corresponds to a unique topological quantum number.

## 3. BH: Buoyancy Harmonics

### Definition

$$U_{g2} = \sum_{m=1}^{\infty} H_m \cdot \left(1 - e^{-[\mathrm{SSq}]\cdot m}\right) \cdot \cos(\omega_{U_{g2}} \cdot t_n)$$

where $H_m = \sum_{k=1}^{m} 1/k$ is the $m$-th harmonic number.

### Saturation Behavior

As $m \to \infty$: $(1 - e^{-[\mathrm{SSq}]\cdot m}) \to 1$

The saturation factor monotonically approaches unity, ensuring $U_{g2}$ is bounded by the harmonic series $\sum H_m \cos(\omega t)$.

### Physical Meaning

BH drives **charge-reactivity gravity** ($U_{g2}$) and saturation harmonics in $E_{\mathrm{net}}(t)$. The harmonic sum $H_m$ introduces logarithmic growth modulated by [SSq]-suppressed saturation and temporal oscillation.

## 4. Cross-Validation

### Internal Consistency

1. **VDS DVP**: DVP samples the VDS polylogarithm at prime arguments. The sum $\sum_p a(p)$ is bounded by $\mathrm{Li}_{26}([\mathrm{SSq}])$.
2. **DVP BH**: Vortex modes (DVP) seed the harmonic structure of $U_{g2}$ (BH) through their topological quantum numbers.
3. **BH VDS**: The vacuum density (VDS) normalizes the saturation parameter in BH, ensuring [SSq] is self-consistent across all three systems.

### Ratio Test

$$\frac{\sum_p a(p)}{\mathrm{VDS}} \ll 1$$

confirms that prime-indexed modes are a sparse subset of the full vacuum series.

## 5. Implementation

Calculator: `VDSDVPBHUnifiedNumberSystemCalculator` in CondensedPhysics.py

- Computes VDS with and without Ramanujan acceleration (configurable $N$ terms)
- Evaluates DVP for the first 15 primes $> 26$
- Computes BH up to harmonic order $M$ with configurable $\omega_{U_{g2}}$ and $t_n$
- Cross-validates DVP/VDS ratio and BH saturation limit

## References


- Ramanujan, S. (1918). On certain trigonometrical sums. *Trans. Camb. Phil. Soc.* 22, 259.
- Murphy, D.T. (2025). UQFF: Star Cradle Mechanics framework. Star-Magic repository.
- PAPER_491: MUGE Compressed Nine-Term Gravity Framework (VDS/DVP/BH canonical definitions).